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Part B:

Question #2:

i) $T(n) = T(n/3) + T(n/3) + n^3$

$T(n) = 2T(n/3) + n^3$

Subtitute: $T(n) = 2[2T(n/9) + (n/3)^3] + n^3$

$T(n) = 4T(n/9) + \frac{2n^3}{27} + n^3$

Expand: $T(n) = 8T(n/27) + n^3\left(1 + \frac{2}{27} + \frac{4}{729}\right)$

$k \rightarrow T(n) = 2^k T(n/3^k) + n^3 \sum_{i=0}^{k-1} \left(\frac{2}{27}\right)^i$

$\Rightarrow \frac{n}{3^k} = 1 \Rightarrow k = \log_3 n \Rightarrow \boxed{T(n) = \Theta(n^3)}$

ii) $T(n) = (n/2)\,T(n/8) + \lg n$; since $\lg n = \log n$

Sub $T(n/8)$: $T(n/8) = \frac{n}{16} T(n/64) + \log(n/8)$

$\Rightarrow T(n) = \frac{n}{2}\left[\frac{n}{16} T(n/64) + \log(n/8)\right] + \log n$

$T(n) = \frac{n^2}{32} T(n/64) + \frac{n}{2}\log(n/8) + \log n$

k: $P_k = \prod_{i=0}^{k-1} \frac{n}{2.8^i}$; $P_k = \frac{n^k}{2^k 8^{k(k-1)/2}}$

$\Rightarrow \frac{n}{8^k} = 1 \rightarrow k = \log_8 n$ || For $n = 8^k$: $P_k = \frac{8^{k(k+1)/2}}{2^k}$

$\Rightarrow T(n) = e^{\Theta((\log n)^2)}$

$\Rightarrow \boxed{T(n) = n^{\Theta(\log n)}}$

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iii) $A_n = A_{n-1} + A_{n-2} + A_{n-3} + \sqrt{n}$

Expand: $A_{n-1} = A_{n-2} + A_{n-3} + A_{n-4} + \sqrt{n-1}$

Sub: $A_n = 2A_{n-2} + 2A_{n-3} + A_{n-4} + \sqrt{n-1} + \sqrt{n}$

Ignore $\sqrt{n}$: $A_n = A_{n-1} + A_{n-2} + A_{n-3}$

Assume $A_n = r^n$: $r^n = r^{n-1} + r^{n-2} + r^{n-3}$

Divide by r^{n-3}: $r^3 = r^2 + r + 1$

$\Rightarrow r^3 - r^2 - r - 1 = 0$

$\Rightarrow r \approx 1.839 \Rightarrow \boxed{A_n = \Theta(1.839^n)}$

iv) $A_n = 3A_{n-n/2} + 2.\lg n$; since $\lg n = \log n$

since $n - n/2 = n/2$

$\rightarrow A(n) = 3A(n/2) + 2\log n$

Expand; $A(n) = 3[3A(n/4) + 2\log(n/2)] + 2\log n$

$A(n) = 9A(n/4) + 6\log(n/2) + 2\log n$

$\Rightarrow A(n) = 3^k A(n/2^k) + 2\sum_{i=0}^{k-1} 3^i \log(n/2^i)$

Stop when $n/2^k = 1 \Rightarrow k = \log_2 n$

$\Rightarrow 3^k = n^{\log_2 3}$

$\Rightarrow \boxed{A(n) = \Theta(n^{\log_2 3})}$

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v) $T(n) = 2T(n/4) + 2T(n-2) + 1$

Expand: $T(n-2) = 2T((n-2)/4) + 2T(n-4) + 1$

Sub: $T(n) = 2T(n/4) + 4T((n-2)/4) + 4T(n-4) + 3$

Observe dom. chain: $T(n) \geq 2T(n-2)$

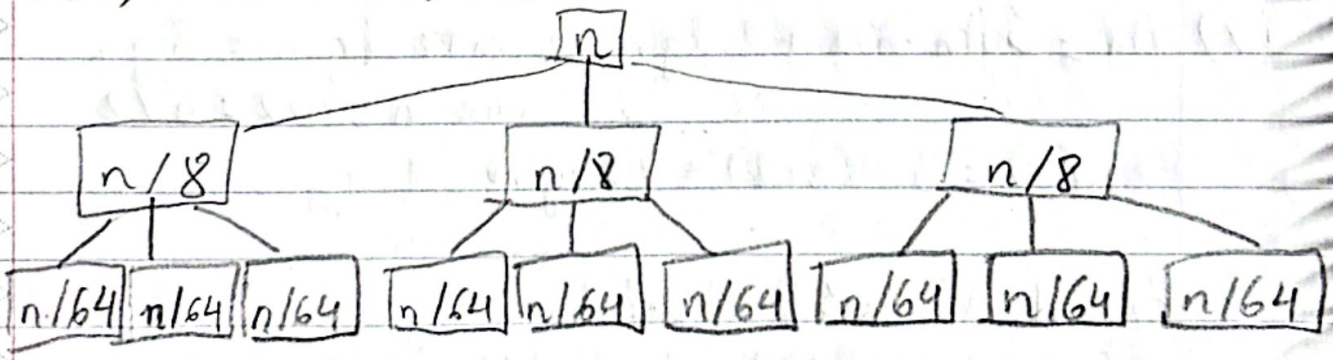
$\rightarrow T(n) = 2^k T(n-2k)$

$\rightarrow T(n) = \Omega(2^{n/2})$

$\Rightarrow$ $T(n) = \Theta(2^{n/2})$ or $T(n) = \Theta((\sqrt{2})^n)$

Question #3:

i) $T(n) = 3T(n/8) + n$



level	# of nodes	Total cost
0	1	n
1	3	$3n/8$
2	9	$9n/64$
i	3^i	$n(3/8)^i$

Height: $h = \log_8 n$

Add the cost: $T(n) = n\left[1 + \frac{3}{8} + \left(\frac{3}{8}\right)^2 + \cdots\right]$

Since $3/8 < 1$, geo. sum is bounded by a constant

$\Rightarrow$ $\boxed{T(n) = \Theta(n)}$

ii) $A_n = 2A_{n/2} + n/2$

Level	Nodes	Cost per node	Total
0	1	$n/2$	$n/2$
1	2	$n/4$	$n/2$
2	4	$n/8$	$n/2$
i	2^i	$n/2^{i+1}$	$n/2$

Height: $h = \log_2 n$

Add the cost: $A(n) = \frac{n}{2}\log_2 n + O(n)$

$\Rightarrow$ $\boxed{A(n) = \Theta(n \log n)}$

iii) $T(n) = T(n/3) + T(n/6) + n$

Level 1: $\frac{n}{3} + \frac{n}{6} = \frac{n}{2}$

Level 2: $\frac{n}{4}$

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→ Pattern: $n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}, \ldots$

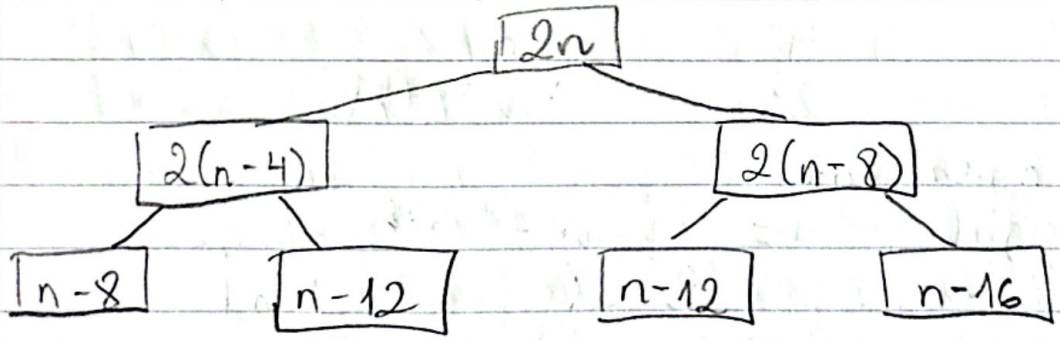
Sum of geo. series:

$$T(n) \leq n\left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) + O(n)$$

Geo. sum is bounded by 2. Also, root alone costs n

$\Rightarrow$ $\boxed{T(n) = \Theta(n)}$

iv) $F_n = F_{n-4} + F_{n-8} + 2n$



Reindex using $n = 4m$

Let: $G(m) = F(4m)$

Then: $G(m) = G(m-1) + G(m-2) + 8m$

Identify tree's growth:

- The recursive branching follows a Fibonacci-type pattern
- The dominant homo. solution satisfies:

$$r^2 = r + 1$$

The largest root is the golden ratio: $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$

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. The forcing term $8m$ is only linear, so the exponential branching dominates for ordinary fixed, non-neg. base cases.

$$G(m) = \Theta(\phi^m)$$

Sub.: $m = n/4$

$$\Rightarrow \boxed{F(n) = \Theta(1.618^{n/4})}$$

v) $2a_{n/6} = a_n - a_{n/2} - n/2$

$$\rightarrow a_n = a_{n/2} + 2a_{n/6} + \frac{n}{2}$$

. Calculate level 0: $C_0 = \frac{n}{2}$

. Calculate level 1:

The children have sizes $n/2$, $n/6$, $n/6$

$$\Rightarrow C_1 = \frac{1}{2}\left(\frac{n}{2} + 2\frac{n}{6}\right)$$

$$C_1 = \frac{n}{2} \cdot \frac{5}{6}$$

. Determine cost ratio: $C_i = \frac{n}{2}\left(\frac{5}{6}\right)^i$

. Add the cost: $T(n) \leq \frac{n}{2} \sum_{i=0}^{\infty} \left(\frac{5}{6}\right)^i + O(n)$

$$= \frac{n}{2} \cdot \frac{1}{1-5/6} + O(n)$$

$$= 3n + O(n)$$

$$\Rightarrow \boxed{a_n = \Theta(n)}$$

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Question #4:

i) $F_n = 4F_{n/4} + n^2$

→ $a=4,\ b=4,\ f(n)=n^2$

. Cal. the critical exp. : $n^{\log_4 4} = n$

. Compare : $n^2 > n$

→ The regularity condition holds:

$$4(n/4)^2 = \frac{n^2}{4}$$

$$\Rightarrow \boxed{F(n) = \Theta(n^2)}$$

ii) $T(n) = 4T(n/8) + n^2/2 \cdot \lg n$

→ $a=4,\ b=8$ $\qquad \lg n = \log n$

$$n^{\log_8 4} = n^{2/3}$$

Interpretation A: $f(n) = \frac{n^2}{2}\log n$; then $f(n)$ grows polynomially faster than $n^{2/3}$

→ Case 3: $\boxed{T(n) = \Theta(n^2 \log n)}$

Interpretation B: if $f(n) = \frac{n^2}{2\log n}$, this also grows polynomially faster than $n^{2/3}$

→ Case 3: $\boxed{T(n) = \Theta\left(\frac{n^2}{\log n}\right)}$

iii) $T(n) = 2T(n/2) + 3T(n/3) + n^2$

→ $n/2$ & $n/3$

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Level 0: $C_0 = n^2$

Level 1: $C_1 = 2(n/2)^2 + 3(n/3)^2$

$$C_1 = \frac{n^2}{2} + \frac{n^2}{3} = \frac{5}{6} n^2$$

$$\Rightarrow C_i = n^2 \left(\frac{5}{6}\right)^i$$

Add the cost: $T(n) \leq n^2 \sum_{i=0}^{\infty} \left(\frac{5}{6}\right)^i + O(n^2)$

Since $5/6 < 1$, sum is bounded by a constant.

The root costs n^2, so: $\boxed{T(n) = \Theta(n^2)}$

iv) $T(n) = 3T(n/4) + \sqrt{n}$

$\rightarrow a = 3,\ b = 4,\ f(n) = n^{1/2}$

$\rightarrow n^{\log_4 3} \approx n^{0.79}$

$\rightarrow n^{0.5} < n^{0.79}$

The additional term is poly. smaller, so Case 1 applies.

$\Rightarrow \boxed{T(n) = \Theta(n^{\log_4 3})}$ or $\boxed{T(n) \approx \Theta(n^{0.79})}$

v) $A_n = 2A_{n-3} + n^3$

. Expand by sub.:

$A_n = 2[2A_{n-6} + (n-3)^3] + n^3$

$A_n = 4A_{n-6} + 2(n-3)^3 + n^3$

$\Rightarrow A_n = 8A_{n-9} + 4(n-6)^3 + 2(n-3)^3 + n^3$

. Find general form:

$$A_n = 2^k A_{n-3k} + \sum_{i=0}^{k-1} 2^i (n-3i)^3$$

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Find the stopping point: $n - 3k = O(1)$

$\Rightarrow k \approx n/3$

$\Rightarrow \boxed{A_n = \Theta(2^{n/3})}$

Question #5:

i) $A_0 = 2,\ A_1 = 0,\ A_n = A_{n-1} + 2A_{n-2}$ for every $n \geq 2$

$A(x) = A_0 + A_1 x + A_2 x^2 + \dots$

$A(x) = 2 + 0x + A_2 x^2 + \dots$

. Multiply recurrence by x^n & sum from $n=2$

$$\sum_{n=2}^{\infty} A_n x^n = \sum_{n=2}^{\infty} A_{n-1} x^n + 2\sum_{n=2}^{\infty} A_{n-2} x^n$$

$\rightarrow A(x) - 2 = x(A(x) - 2) + 2x^2 A(x)$

$A(x)(1 - x - 2x^2) = 2 - 2x$

$$A(x) = \frac{2 - 2x}{1 - x - 2x^2}$$

$\Rightarrow 1 - x - 2x^2 = (1 - 2x)(1 + x)$

$$\Rightarrow A(x) = \frac{2/3}{1 - 2x} + \frac{4/3}{1 + x}$$

. Extract coef. of x^n:

$$\frac{1}{1 - 2x} = \sum_{n=0}^{\infty} 2^n x^n$$

$$\frac{1}{1 + x} = \sum_{n=0}^{\infty} (-1)^n x^n$$

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$$\Rightarrow \boxed{A_n = \frac{2}{3}(2^n) + \frac{4}{3}(-1)^n}$$

Check:

. $n=0$: $A_0 = \frac{2}{3} + \frac{4}{3} = 2$ ✓

. $n=1$: $A_1 = \frac{4}{3} - \frac{4}{3} = 0$ ✓

ii) $T(0)=0, T(1)=1, T(n)=2T(2n-2)+1$ for every $n \geq 2$

. Define OGF: $T(x) = \sum_{n=0}^{\infty} T(n)x^n$

. Multiply by x^n & sum:

$$\sum_{n=2}^{\infty} T(n)x^n = 2\sum_{n=2}^{\infty} T(n-2)x^n + \sum_{n=2}^{\infty} x^n$$

. Sub. OGF expressions:

$$T(x) - x = 2x^2 T(x) + \frac{x^2}{1-x}$$

$$\Rightarrow T(x)(1-2x^2) = x + \frac{x^2}{1-x}$$

$$T(x) = \frac{x}{(1-x)(1-2x^2)}$$

. Find the sequence:

n	$T(n)$
0	0
1	1
2	1

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3	3
4	3
5	7
6	7
7	15

$\Rightarrow T(2k) = 2^k - 1$

$\Rightarrow T(2k+1) = 2^{k+1} - 1$

$\Rightarrow \boxed{T(n) = 2^{\lceil n/2 \rceil} - 1}$

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